

Food Delivery

Req2Design · IEEE 830-1998 aligned

Document Control

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1:Introduction

1.1:Purpose

Purpose: To provide a detailed specification for the development of a food delivery application that enables customers, restaurants, and drivers to interact seamlessly throughout the ordering process.

1.2:Scope

Scope: This document specifies the requirements for a food delivery application focusing on core functionalities including browsing, customization, scheduling, tracking, order confirmation, payment handling, and notification systems. It does not cover advanced features such as security measures, scalability, or maintenance procedures.

1.3:Definitions/Acronyms

- **API:** Application Programming Interface
- **GUI:** Graphical User Interface

1.4:References

References: IEEE Std 830-1998

1.5:Overview

Overview: The food delivery application will facilitate efficient order management from customer request through driver delivery, ensuring timely and accurate service.

2:Overall Description

2.1:Product Perspective

Product Perspective: The application aims to create a user-friendly interface for both customers and businesses involved in the food delivery ecosystem. It integrates various services such as restaurant listings, order customization options, real-time tracking capabilities, and automated payment processing.

2.2:Product Functions

- Customer functions include browsing menus, customizing orders, setting up delivery schedules, and receiving updates on their orders' statuses.
- Restaurant functions encompass viewing incoming orders, confirming preparation times, marking items as ready, and pausing orders temporarily under certain conditions.
- Driver functions involve accepting delivery assignments, following optimal route directions, and reporting any issues encountered during transit.

2.3:User Characteristics

User Characteristics: Primary users consist of customers who place food orders, restaurants that prepare meals, and drivers responsible for delivering these orders. Secondary users might include administrative staff managing backend operations.

2.4:Constraints

Constraints: The application must operate within standard internet protocols without requiring specialized equipment beyond typical mobile devices and web browsers. Timeouts and retries for network failures must be implemented to ensure robustness.

2.5:Assumptions/Dependencies

Assumptions/Dependencies: All parties have access to reliable internet connections; third-party APIs may be utilized for mapping services and payment gateways.

3:Specific Requirements

3.1:External Interface Requirements

- **User Interface:** A responsive design supporting desktop and mobile platforms, featuring intuitive navigation and interactive elements.
- **Hardware Interface:** Compatibility with common smartphones and tablets running iOS and Android operating systems.
- **Software Interface:** Integration with popular web browsers and native applications.
- **Communication Interface:** Secure transmission of data via HTTPS protocol, adhering to industry standards for encryption and authentication.

3.2:Functional Requirements

FR-01 Browse Restaurants

Input: None

Processing: Display list of registered restaurants along with their cuisines and ratings.

Output: List of restaurants with relevant details.

Priority: High

FR-02 Customize Orders

Input: Menu items selected by the customer.

Processing: Allow modifications to quantities, add-ons, and special requests.

Output: Updated order summary reflecting changes made by the customer.

Priority: Medium

FR-03 Schedule Deliveries

Input: Customer's preferred delivery date and time.

Processing: Validate availability based on restaurant capacity and current rider workload.

Output: Confirmation message indicating whether the requested slot is available.

Priority: Low

FR-04 Track Riders Live

Input: Order ID provided by the customer.

Processing: Retrieve location updates from the assigned driver's device.

Output: Real-time map showing the driver's progress towards the destination.

Priority: High

FR-05 Confirm Preparation Times

Input: Order ID sent to the restaurant.

Processing: Receive estimated completion time from the restaurant kitchen.

Output: Notification to the customer about when their order will be ready.

Priority: Medium

FR-06 Mark Orders Ready

Input: Order ID communicated between the restaurant and the system.

Processing: Update the status of the order to 'ready'.

Output: Confirmation feedback displayed to the restaurant staff.

Priority: Medium

FR-07 Accept Delivery Tasks

Input: Delivery task assignment received from the system.

Processing: Verify vehicle readiness and start navigation.

Output: Route guidance information presented to the driver.

Priority: High

FR-08 Navigate Optimized Routes

Input: Current location of the driver and final delivery address.

Processing: Calculate shortest path considering traffic conditions and other active deliveries.

Output: Detailed driving directions and estimated travel time.

Priority: High

FR-09 Report Incidents

Input: Incident type and description entered by the driver.

Processing: Log incident details and notify relevant personnel.

Output: Acknowledgment message confirming receipt of the report.

Priority: Medium

FR-10 Retry Failed Payments

Input: Payment failure indication from the payment gateway.

Processing: Attempt reprocessing transaction after a predefined timeout period.

Output: Success or failure result of the retry attempt.

Priority: Medium

FR-11 Handle Order Substitutions

Input: Requested substitution details from either party.

Processing: Modify existing order record according to new specifications.

Output: Updated order information reflected across all interfaces.

Priority: Medium

Non-Functional Requirements

Usability: The application shall support easy-to-use interfaces tailored to different skill levels and preferences, ensuring accessibility for all users.

Reliability: The system must maintain consistent operation even under heavy load scenarios, preventing crashes due to unexpected inputs or external disruptions.

Performance: Response times for critical actions such as placing an order, updating status, and navigating routes shall remain under 5 seconds.

Portability: Codebase shall be designed to adapt easily to future platform expansions while preserving functionality and performance characteristics.

4: System Features

Feature 1: Browsing Menus

- Customers can view menus offered by participating restaurants.
- Includes filtering options based on cuisine types and dietary restrictions.

Feature 2: Customization Options

- Allows customers to modify portion sizes, select additional toppings, and specify any unique requirements.

Feature 3: Real-Time Tracking

- Provides ongoing updates on the whereabouts of delivered goods, allowing customers to anticipate arrival times accurately.

Feature 4: Automated Notifications

- Sends push alerts regarding significant milestones in the order lifecycle, enhancing transparency and convenience.

Feature 5: Efficient Routing Algorithms

- Utilizes sophisticated algorithms to determine optimal paths for drivers, minimizing overall transportation costs and reducing wait times for customers.

Feature 6: Seamless Integration

- Ensures smooth interaction among diverse components like front-end displays, back-end databases, and external service providers.

End of Document.